

Research Proposal

An Uncertainty-Aware Pipeline for Automated Scientific Knowledge Extraction and Causal Reliability Assessment: A Cross-Domain Validation on Boron-Doped Diamond and Polymeric Membrane Literature

1. Background and Research Gap

The exponential growth of scientific literature has created a bottleneck in evidence synthesis: the accumulated knowledge in any specialized field now exceeds what any individual researcher can manually integrate. Automated information extraction using Large Language Models (LLMs) has emerged as a response, with systems such as MaTableGPT demonstrating that structured tabular data can be extracted from materials science literature at F1 scores approaching 97% in domains like water-splitting catalysis. Parallel work has extended extraction to multi-element tabular data in polymer property literature (Phi et al., ACL 2025).

However, three limitations persist across this body of work:

First, existing systems evaluate extraction quality in isolation, typically via precision, recall, or F1 against a gold-standard subset without any reference to how extracted data will be used downstream. An extraction error that is inconsequential for one analytical task may be catastrophic for another, yet no existing framework quantifies this task-dependent impact.

Second, no existing system propagates extraction uncertainty into downstream inference. Extracted values are treated as ground truth once written to the database, even though the extraction process itself may yield different values under different prompt formulations, and even though some values are inherently more ambiguous in the source text than others. The reliability of any downstream statistical or causal claim is therefore disconnected from the reliability of its inputs.

Third, the dominant application domain of these systems - structured tabular data - excludes a substantial fraction of scientific knowledge that appears only in narrative prose. Performance values, service lifetimes, and failure thresholds are frequently reported inside paragraphs rather than tables, and this prose-embedded data remains largely inaccessible to current automated extraction pipelines.

This proposal addresses all three limitations through a single integrated framework: an **uncertainty-aware extraction-to-causal pipeline** in which (i) extraction confidence is

quantified per value via self-consistency sampling under paraphrased prompts, (ii) this confidence is propagated into causal structure learning via confidence-weighted bootstrap resampling, and (iii) extraction quality is evaluated not by F1 alone but by its **causal impact** i.e the degree to which correcting an extraction error changes the resulting causal graph. The framework is validated across two deliberately dissimilar domains: boron-doped diamond (BDD) electrodes and polymeric membranes (PVDF/hydroxyapatite), to test whether the uncertainty-reliability relationship is domain-dependent.

2. Research Questions

1. Can extraction uncertainty, quantified via self-consistency sampling under paraphrased prompts, be computed reliably for both numerical and categorical variables extracted from narrative scientific text?
2. Does extraction uncertainty correlate with extraction error, as measured against a manually-verified gold standard? (Validity check for the uncertainty metric itself.)
3. Does extraction uncertainty propagate into causal structure instability, as measured by (a) Structural Hamming Distance between causal graphs learned from original versus corrected extractions, and (b) the statistical association between per-record extraction uncertainty and per-edge bootstrap stability?
4. Does the strength of this uncertainty-instability relationship differ between BDD electrochemistry and polymeric membrane literature, and if so, can the difference be attributed to identifiable structural properties of the data (variable density, unit homogeneity, terminological consistency)?

3. Methodology

Stage 1: Literature Search and Data Compilation

Domain 1 (BDD electrochemistry): Structured search across accessible databases (2015–2026) using keyword strings combining "boron-doped diamond" with "Raman," "delamination," "Faradaic efficiency," "critical current density," "service life," and "substrate." Inclusion criteria: peer-reviewed studies reporting (i) Raman-derived doping or sp^3/sp^2 data AND (ii) at least one quantitative performance or failure outcome, for a named substrate (Ti, Nb, Ta, Si).

Domain 2 (PVDF/HAp membranes): Parallel search using "PVDF," "hydroxyapatite," "membrane," combined with "flux," "rejection rate," "porosity," "FTIR," "XRD." Inclusion criteria: peer-reviewed studies reporting (i) at least two characterization modalities (spectroscopic or diffraction) AND (ii) at least one quantitative performance outcome.

Extraction targets: For each included study, a predefined JSON schema specifies variables to extract (numeric: doping level, thickness, current density, service life, flux, rejection rate; categorical: substrate type, deposition method, membrane composition; relational: reported associations between variables). Schema is domain-specific but shares a common structure to enable cross-domain comparison.

Stage 2: Uncertainty-Aware Extraction

For each target value, the extraction query is executed $N = 7\text{--}10$ times under **paraphrased prompts** (semantically equivalent reformulations, not literal repetitions), using a fixed LLM with controlled temperature.

Uncertainty quantification:

- **Numeric variables:** coefficient of variation (CV) across N attempts, computed after outlier handling (values beyond $3\times$ median absolute deviation from the median are excluded, and the exclusion rate is reported).
- **Categorical/relational variables:** $1 - (\text{majority agreement proportion})$ across N attempts.

Validity check (gold standard): A manually-verified subset of 100–150 extracted values per domain is created by the researcher, with ambiguous cases resolved by reference to the source text. Extraction precision, recall, and F1 are computed against this subset. The correlation between per-value uncertainty and per-value extraction error is then measured, this is the validity check for the uncertainty metric itself. If uncertainty does not correlate with error, the metric is not valid and the pipeline's central claim fails (this is explicitly stated as a possible outcome).

Stage 3: Causal Structure Learning with Uncertainty Propagation

Baseline causal learning: The PC algorithm (via causal-learn) is applied to the extracted dataset, with variable selection guided by domain knowledge to keep the variable set small (target: 5–8 variables per domain). Given the anticipated sample size (15–25 records per domain), the PC algorithm is used **not as a definitive structure discovery method** but as a **structure proposal mechanism** whose sensitivity to input uncertainty is the object of study. This limitation is stated explicitly.

Uncertainty propagation: Confidence-weighted bootstrap resampling is applied, where each record's sampling weight is defined as $w_i = 1 / (1 + CV_i)$ for numeric-dominated records, or $w_i = \text{agreement proportion}$ for categorical-dominated records. The exact weighting function is reported explicitly, and alternative weighting functions (linear, exponential) are tested in sensitivity analysis.

Structure stability metrics:

- **Structural Sensitivity:** Structural Hamming Distance (SHD) between the graph learned from original extractions and the graph learned after selectively correcting high-uncertainty extractions.
- **Edge-Stability Sensitivity:** For each edge, bootstrap stability is computed as the frequency of edge appearance across resamples. The association between per-record extraction uncertainty and per-edge stability is then measured via correlation, with confidence intervals.

Stage 4: Cross-Domain Comparison

The full pipeline is run independently on Domain 1 (BDD) and Domain 2 (PVDF/HAp). The following are compared:

- Distribution of extraction uncertainty across domains.
- Correlation between uncertainty and error (validity check) across domains.
- Strength of uncertainty-instability relationship across domains.
- Structural properties of each dataset (variable density, unit homogeneity, terminological consistency) that may explain differences.

Stage 5: Pre-Registered Hypotheses

Three hypotheses are registered before any implementation:

H1 (Substantial effect): If high-uncertainty records are statistically associated with unstable edges → the contribution is that uncertainty propagation is a necessary, not optional, step in causal inference pipelines built on automatically extracted data.

H2 (Negligible effect): If no significant association is found → the contribution is that causal inference on this class of extracted data is robust to extraction noise within the observed range, a reassurance result, not a failure.

H3 (Domain-dependent effect): If the relationship strength differs between domains → the contribution is a diagnostic criterion identifying when uncertainty propagation is necessary based on structural data properties.

Acceptance criteria: For H1 and H3, a minimum effect size of $|r| > 0.3$ with a 95% confidence interval excluding zero is required. For H2, a null result is accepted only if the confidence interval for the correlation excludes $|r| > 0.3$ (i.e., the absence of a substantial effect is itself statistically supported).

4. Novelty and Contribution

Methodological contributions:

1. **Task-aware extraction evaluation:** A new evaluation metric - causal impact of extraction error - that measures extraction quality by its effect on downstream inference rather than in isolation.
2. **Uncertainty propagation through a multi-stage pipeline:** A framework that carries extraction confidence into causal structure learning, with explicit mathematical definitions and testable validity checks.
3. **Cross-domain validation:** Empirical evidence on whether the uncertainty-reliability relationship is domain-dependent, with a diagnostic criterion for when uncertainty propagation is necessary.

Empirical contributions:

1. A manually-verified gold-standard dataset for extraction from BDD and PVDF/HAp literature.
2. Open-source implementation of the full pipeline.

3. Transparent reporting of negative or null results if the central hypothesis is not supported.

5. Expected Outcomes

- A validated uncertainty metric for LLM-based scientific extraction, with quantified correlation to extraction error.
- A quantified relationship between extraction uncertainty and causal structure instability, with domain comparison.
- An open-source implementation of the pipeline.
- A reusable gold-standard dataset for future benchmarking.
- Explicit reporting of the conditions under which uncertainty propagation is necessary versus unnecessary.

6. Explicit Limitations

- **Sample size:** Causal structure learning with 15–25 records per domain cannot produce definitive causal graphs. The study's claim is about **sensitivity of proposed structures to input uncertainty**, not about discovering true causal structure.
- **Uncertainty type:** Only extraction uncertainty is measured. Aleatoric uncertainty (source text ambiguity) and epistemic uncertainty (model knowledge gaps) are acknowledged as distinct and left to future work requiring separate methodology.
- **Domain coverage:** Two domains are tested. Generalization to other scientific fields is a hypothesis, not a demonstrated result.
- **Gold-standard subjectivity:** Manual verification introduces researcher judgment. Inter-rater reliability is not assessed in this proposal; this is a limitation.
- **LLM version dependence:** Results may vary with LLM version. The specific model and version are reported, and version-dependence is acknowledged.

7. Target Venues

Primary:

- *ACL* or *EMNLP* (NLP for scientific information extraction)
- *NeurIPS Datasets and Benchmarks Track* (methodological contribution with benchmark)
- *AISTATS* or *UAI* (causal inference methodology)

Secondary:

- *Journal of Chemical Information and Modeling* (AI for chemistry)
- *Patterns* (Cell Press) (AI for science, cross-domain methodology)

8. Keywords

scientific information extraction; large language models; uncertainty quantification; causal discovery; self-consistency sampling; cross-domain validation; boron-doped diamond; polymeric membranes

9. Feasibility Statement

All components are implementable with open-source tools (Python, causal-learn, numpy, pandas) and LLM API access. No specialized hardware, paid software, or laboratory access is required. The primary feasibility constraint is the availability of sufficient usable records after gold-standard verification; this is checked empirically during Stage 1 before full pipeline implementation. If Domain 1 or Domain 2 yields fewer than 15 usable records, the study is reframed as a single-domain methodological study with the second domain used as a reduced-scope transfer test, and this decision is made before Stage 3 begins.

10. Plan B

If the uncertainty-error correlation (validity check) is not supported, the study is reframed as a **negative result on the validity of self-consistency uncertainty metrics for scientific extraction**, which is itself a publishable contribution, since it would caution the community against adopting such metrics without validation. The pipeline implementation and gold-standard dataset remain as contributions independent of the central hypothesis outcome.

End of Proposal